

A course in language creation



Conlangs University

Phonology 3

Distinctive features & rules notation



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Hello again Conlangs U!

In this lesson, we tackle how to express the rules that dictate how your phonemes and sounds interact with each other.

"How is this useful?" you may ask, and you would be right to: after all, this isn't as much creation as it is a rather boring task of analysis... to some. I personally enjoy it a lot.

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If you wish to learn the features, there is a piece of software dedicated to this from the UCLA: [Pheatures](#).

Prerequisites

You will need to have read the first two parts of this series on Phonology and understand how speech is produced:

- what places of articulation are, and what the vowel space is
- how the vocal folds vibrate
- what manners of articulation are

This lesson will also give you the elements you will need for the next one, on sound changes.

Distinctive features

If your sound changes regularly affect, or are regularly affected by, a particular type of sounds, **distinctive features** can be used to group them into **Natural Classes**.

NOTE This section mainly follows the theories from

- *Preliminaries to Speech Analysis*, by Roman Jakobson, Gunnar Fant & Morris Halle (1952);
- *The Sound Pattern of English*, by Noam Chomsky & Morris Halle (1968).

You can find a short summary of the following [at this link](#).

While the *phoneme* is the smallest meaning distinguishing unit within a word, there is a unit smaller than it — though it is not necessarily meaningfully contrastive.

This unit is known as the *distinctive feature*.

These features are phonetically-based properties (which can be acoustic, articulatory, or both) used by phonological analysis and, while a lot of them can be notated using the IPA, we will go over a notation for them specifically. They are used to define not only the possible sounds of human languages, but also phonological rules.

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Each feature can be defined by a binary value of *plus* + or *minus* -, meaning that for each speech sound, the segment either has ([+feature]) or does not have ([-feature]) the feature.

The binarity of these properties is often only relevant *phonologically*, while it is not always appropriate for a *phonetic analysis*.

For example, the feature [± nasal] is binary phonologically, but phonetically a speech sound can be nasalised to different degrees: while the phonemic contrast is generally whether or not the sound is nasalised, how much the [+ nasal] sound is nasalised might vary.

Some of those features involve relative movement of the tongue. The point of reference used for this is that of the tongue at rest, roughly equivalent to the position of the tongue during the production of the vowel [ə].

Major class features

There are 3 major class features: [± **consonantal**], [± **syllabic**], and [± **sonorant**].

[+ **consonantal**] means that there is a major obstruction of the vocal tract happening in the oral cavity.

[-**consonantal**] means there isn't: it is an attribute of vowels, which feature a maximally open vocal tract.

[± **syllabic**] is determined by whether [+ **syllabic**] or not [-**syllabic**] a sound can be the nucleus of a syllable.

[± **sonorant**] is dependent on the sound being produced with a vocal tract configuration that allows for spontaneous voicing. In simpler words, a sound that can be hummed or sung is typically a sonorant.

[+ **sonorant**] sounds are louder than [-**sonorant**] ones, but it is hard to define an exact and consistent cutoff point.

Here is a quick chart to help you remember the broad strokes of this classification:

↓ Features	Obstruents (s, z, p, b)	Glides (j, w)	Nasals (n, m), liquids (r, l)	Vowels (a, i, u)	Syllabic consonants (ɾ, ɭ, ɹ)
[± consonantal]	+	-	+	-	+
[± syllabic]	-	-	-	+	+
[± sonorant]	-	+	+	+	+

Classes defined by more than one feature have their features written between the same set of brackets. As such, **obstruents** would be written as [+consonantal -syllabic -sonorant] . Abbreviating the name of the features is oft done for ease of reading: [+cons -syl +son] .

Conversely, [+cons] [-syl] refers to two classes, one defined by [+consonantal] and the other by [-syllabic].

It can also be noted that all [+syllabic] sounds are also [+sonorant], as syllabic obstruent (which would be [+cons +sylv -son]) are unattested.

Another impossible combination is [-cons -son] , as it is physically unfeasible: to be [-sonorant] segment would need a major obstruction of the vocal tract, but [-consonant] means that there cannot be an obstruction in the oral cavity. This would only leave a constriction of the nostrils, which is not feasible.

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Nasals and liquids could be separated by the addition of a fourth feature [± **approximant**] . With it, vowels, glides and liquids would be [+ **approximant**] and nasals and obstruents would be [- **approximant**] .

Those four features give us an approximation of the *sonority hierarchy* that was talked about in the previous Phonology lesson:

↓ Features	Vowels	Glides	Liquids	Nasals	Obstruents
[± consonantal]	-	-	+	+	+
[± syllabic]	+	-	-	-	-
[± approximant]	+	+	+	-	-
[± sonorant]	+	+	+	+	-

Features of manner

There are 4 features that describe **manner of articulation**: [$\pm$ **continuant**], [$\pm$ **delayed release**], [$\pm$ **nasal**], and [$\pm$ **lateral**].

See **Manner of articulation** in CU Phonology 1.

[**+continuant**] sounds are those with a continuous airflow, without a complete obstruction of the oral cavity.

[**+delayed release**] (or [**+del.rel**] or [**+DR**]) sounds are those where the tongue does not immediately release total constriction of the vocal tract, so a fricative sound is realised after a stop.

[**+nasal**] sounds are those that have the air pass through the nasal cavity by lowering the velum.

[**+lateral**] sounds have the tongue's middle section lowered at the side, allowing the air to flow laterally. Most [**+lateral**] sounds are *coronal*.

Here are some nominal examples, using sounds at an alveolar place of articulation:

↓ Features	t, d	s, z	tʃ, dʒ	m, n	l	ɭ
[$\pm$ continuant]	-	+	-	-	+	+
[$\pm$ delayed release]	-	-	+	-	-	-
[$\pm$ nasal]	-	-	-	+	-	+
[$\pm$ lateral]	-	-	-	-	+	+

A sound cannot be both [**+lat -cont**], as a lateral release is still a release and does not fully obstruct the vocal tract.

tʃ however has to be considered an exception to this rule as its only distinction from tθ, in terms of features, is its laterality. **Affricate consonants** are necessarily [**-cont**] as they start with a plosive, which completely block the airflow, and instead are described as [**+delayed release**].

It is to be noted that defining a sound as being [**+continuant**] is **not the same** as saying it is a fricative: although fricatives are [**+continuant**], the sounds commonly associated with that label are also [**-sonorant**].

"Continuants" as a label groups together vowels, glides, and fricatives.

Nasal consonants are [**-continuant**]: even though the airflow is not completely stopped and redirected through the nose, they involve a total closure of the oral cavity.

Nasal vowels are of course [**+nasal**] and [**+continuant**].

Laryngeal features

Laryngeal features include [±**voice**], [±**spread glottis**] or [±s.g.], and [±**constricted glottis**] or [±c.g].

Voiced sounds are notated [+**voice**] and unvoiced sounds are notated [-**voiced**].

[±**spread glottis**] describes aspiration in obstruents (**p^h**) and breathiness in sonorants (**m**, **n**). It means that the vocal cords are set further apart, opening the glottis.

[±**constricted glottis**] describes implosives (**b**, **ɟ**), ejective obstruents (**p'**) and laryngealisation (creaky voice) in sonorants (**ɹ**, **ŋ**). The glottal stop also makes use of this. It means that the glottis is narrow or closed.

The two features for the glottis' spread or constriction allow for a three-way distinction for glottal width or openness.

The majority of sounds however use what we could call "normal" phonation, which is characterised by [-**s.g** -**c.g**].

↓ Features	p	p^h	p'	b	b^h	ɓ
[± voice]	-	-	-	+	+	+
[± s.g.]	-	+	-	-	+	-
[± c.g.]	-	-	+	-	-	+

The lack of distinction between implosives and ejectives can be a bother in some cases, and while it can be said that implosives are [+voice] while ejectives are [-voice], a problem remains in that implosives do not typically follow the same patterns as other [+voice] segments in the phonological systems of languages. Furthermore, some languages (such as **Lendu**) are analysed as having both voiced and unvoiced implosives.

As such, a number of linguists use an additional [±**implosive**] specifically for those.

Features of place

The features that describe where in the mouth sounds are articulated are as follows: [±**low**], [±**high**], [±**back**], [±**front**], [±**round**], [±**tense**], [±**advanced tongue root**] or [±ATR], [±**anterior**], [±**coronal**], [±**distributed**], and [±**strident**].

While most of them apply to both vowels and consonants, some ([±tense] and [±ATR]) only apply to vowels, while the features [±anterior], [±coronal], [±distributed], and [±strident] only apply to consonants.

Vowel place features

[+**back**] sounds are produced with the body of tongue moved back from the neutral position.

[+**front**] sounds are produced with the body of tongue moved forward from the neutral position.

Here, we use two features where some phonologists would only use one [+**back**] .

This system with both [+**front**] and [+**back**] allows a three-way distinction in front ([+**front** -**back**]), central ([-**front** -**back**]) and back ([-**front** +**back**]) vowels, while one with only [+**back**] necessitates treating central vowels as a type of back vowel.

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[+**low**] sounds are produced with the body of tongue lowered, comparatively to a neutral position.

[+**high**] sounds are produced with the body of tongue raised, comparatively to a neutral position.

NOTE

While [+**high** +**low**] is an impossible configuration, [-**high** -**low**] is not: this only means the tongue is in a neutral position. As such, high vowels are [+**high** -**low**] , low vowels are [-**high** +**low**] , and mid vowels are [-**high** -**low**] .

Just like features for backness/frontness described above, the two features for height allow for a three-way distinction between high, mid and low vowels wherein high vowels are [+**high** -**low**] , mid vowels are [-**high** -**low**] , and low vowels are [-**high** +**low**] .

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[+**tense**] means the sound requires a deliberate muscular effort more intense than [-**tense**] sounds. A non-tense vowel is said to be **lax**.

[+**tense**] vowels are generally more "extreme" in height and front-or-backness than their [-**tense**] counterparts: [+**tense**] front vowels are typically higher and more front, and [+**tense**] back vowels are typically higher and more back.

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[+**round**] sounds have the lips protruded and forming a rounder shape.

No language is known to distinguish more than two degrees of rounding phonemically. However, this does not mean that two sounds distinguished by roundedness in a given language are on extreme ends of "completely rounded" and "completely unrounded", and the exact degrees of roundedness being acknowledged as distinctive can vary greatly.

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[±**ATR**] means the tongue root is drawn forward.

NOTE

The existence of a phonetic distinction between [±**tense**] and [±**ATR**] is subject to debate, and seemingly only really relevant in some languages (such as [Maasai](#) and [Akan](#)) as a way to explain the mechanisms of vocalic harmony.

Sadly, only so much research has been done pertaining to the phonetics of these languages, and the data is not enough to be conclusive.

Here is a summary of the features for vowels:

↓ Height / Front/backness →	[+ front -back]	[-front -back]	[-front + back]
[+ high -low + tense]	i y	ɨ ʉ	ɯ u
[+ high -low -tense]	ɪ ʏ	- -	- ʊ
[-high -low + tense]	e ø	ə ɵ	ɤ ɔ
[-high -low -tense]	ɛ œ	ə ɐ	ʌ ɔ
[-high + low + tense]	- -	ɐ -	- -
[+ high + low -tense]	æ œ	a -	ɑ ɒ

As you can see, I chose to represent them in a very similar layout to the one of the IPA vowel chart: horizontal axis for backness, vertical axis for openness/height, and unrounded phones on the left and rounded ones on the right in each cell.

You may notice however that [a] is noted as central, which is a convention many linguists follow as a distinction between the lowest front vowel possible and a slightly higher one is often unnecessary, and the central low vowel seems to be the most common of all across all languages.

Consonant place features

[+ **anterior**] sounds are those produced with a major constriction or obstruction at the front of the oral cavity (at the alveola or before).

By definition, labials, labiodentals, dentals and alveolars are [+ anterior].

[s] and [ʃ] are differentiated as [s], being alveolar, is [+ anterior] and [ʃ], a post-alveolar, is [-anterior].

[+ **coronal**] sounds are produced with the tongue raised towards the roof of the mouth.

[+ **distributed**] sounds are produced with a constriction over several places of articulation. It applies to coronal consonants.

[+ **strident**] sounds are noisier than their [-**strident**] counterparts.

NOTE

[± **strident**] as a feature is not necessary to apply to all consonants: it is best used comparatively, adding it where *needed* to distinguish between two fricatives that would otherwise have the same feature set. For instance, it is useful to distinguish between [f s] ([+ strident]) and [ɸ θ] ([- strident]).

Using the first three features, it gives us these distinctions for each main place of articulation, using sounds that are [+cons -syl -voi -cont -nas] :

↓ Features / Consonants →	p	t̪	t	tʃ	t	c, k, q, ʃ, ʔ
[± anterior]	+	+	+	-	-	-
[± coronal]	-	+	+	+	+	-
[± distributed]	0	+	-	+	-	-

So, what can we see here?

- consonants articulated at the alveola or fronter are indeed marked as [+ anterior];
- as p does not
- all of c, k, q, ʃ, ʔ are not distinguished.

Waiiit wait wait wait, what does that 0 mean?

Consider the consonant to which that 0 is applied: it cannot be [+ distributed] as it only involves a single point of articulation: the lips. In the table below, I marked as 0 all the features that are either impossible or irrelevant to the distinctions.

In order to distinguish the back consonants, we need to add a few features that were previously introduced as pertaining to vowels:

↓ Features / Consonants →	p	t̪	t	tʃ	t	c	k	q	ʃ	ʔ
[± anterior]	+	+	+	-	-	-	-	-	-	-
[± coronal]	-	+	+	+	+	-	-	-	-	-
[± distributed]	0	+	-	+	-	0	0	0	0	0
[± front]	0	0	0	0	0	+	0	-	-	0
[± back]	0	0	0	0	0	-	+	+	+	0
[± high]	0	0	0	0	0	+	+	-	-	0
[± low]	0	0	0	0	0	-	-	-	+	0

Those 0 again!

Indeed! Consider again p for instance: the distinctions between front, back, high, and low do not matter here.

Had it been p^j (palatalised), then the feature [+high] would have been relevant to include, as palatalisation involves the tongue moving upwards. However, a plain p does not require these distinctions, as the position of the tongue during a sound that does not involve it as an articulator seems to be *whatever is most convenient for articulation, depending on previous and next sounds* and often falls on whatever position it was in prior to articulating [p].

We can notate this as [0high 0low] and so on.

Secondary articulations

[±**round**] can be used for distinguishing between sound produced with rounded or pursed lips, by contrast with [**-round**] for those produced with flat lips.

Labialisation adds [+**anterior** +**round**] .

Palatalisation always invokes the use of [+**high**] , as palatalisation involves raising the tongue closer to the roof of the mouth.

Velarisation involves retracting the tongue towards the velum, and adds [+**high** +**back**] .

Pharyngealisation adds [+**back** +**low**] , as the tongue retracts towards the pharynx.

Dentalisation means an articulator extends to the teeth. It can be expressed by the addition of [+**anterior** +**distributed**] .

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Some linguists add a feature [±**labial**] to this set, which can be useful for labialisation distinctions specifically.

Prosodic features

There are two features describing prosody: [±**stress**] and [±**long**] . Their roles should be fairly obvious from their names:

[+**stress**] sounds are emphatically produced, with greater amplitude and/or higher pitch.

[+**long**] sounds are longer than their [**-long**] counterparts. Long vowel are, of course, [+**long**] .

Trills, taps, & flaps

Nothing in the features described distinguishes trills, taps, and flaps.

To remedy this, phonologists have used [±**trill**] and [±**tap**] , the latter serving for both taps and flaps with the added precision that typically taps are alveolar, thus [+**anterior**] , while flaps are retroflex, [**-anterior**] .

Phonological rules

Here is what a basic phonological rule looks like:

$$A \rightarrow B / _Z$$

It can be read as "*A turns into B in front of Z*" or "*If A occurs before Z, A turns into B*". Let's analyse how this works:

- The arrow, $\rightarrow$ or $>$, indicates the direction of the change. Here, A turns into B;
- The slash $/$ introduces the environment in which the change takes place;
- The underline $_$ indicates the position of the input (A) in relation to other phonological elements. Here, it is positioned *before Z*.

Of all those parts, the environment is optional if the change is not conditioned by it.

If you wanted to express the rule "*A turns into B after Z*", you would notate it this way:

$$A \rightarrow B / Z_$$

If you wanted to write a sound change happening between two different sounds, you would write it as:

$$A \rightarrow B / Y_Z$$

This means that the sound A turns into B when preceded by Y and followed by Z.

NOTE

While both $A \rightarrow B$ and $A > B$ have the same meaning of "sound A turns into sound B", usage has it that $\rightarrow$ is for synchronic change while $>$ is used for diachronic change.

[Wikipedia article about synchrony and diachrony](#)



Some cases are not possible to define using just sounds. You might want/need to signify that a change occurs at the end of a syllable or at the beginning of a word.

And what about groups of sounds, or no sound at all?

Boundaries & composition

There exist symbols commonly used for some of those cases. Here are for instance two rules meaning, respectively, "A turns into B at the end of a word" and "A turns into B at the beginning of a word":

```
A → B / _#  
A → B / #_
```

Another frequent environment is syllable boundary, which is notated in multiple ways. For your documentation and for the sake of consistency, you should probably pick one and stick to it. Here are several ways one can represent sound changes occurring at syllable boundaries:

```
A → B / _$  
A → B / _%  
A → B / _]σ
```

All three rules above read the same way: "A turns into B at the end of a syllable".

In the third rule, we find the symbol Sigma σ, which indicates a syllable. The left-facing bracket indicates where the syllable boundary takes place.

Here are notations, using the same symbols, for the rule "A turns into B at the **beginning** of a syllable":

```
A → B / $_  
A → B / σ[_
```

The Sigma σ can be used without the bracket, representing a syllable:

```
A → B / σ_  
A → B / _σ
```

The two rules above can be read, respectively, as "A turns into B after a syllable" and "A turns into B before a syllable".

Note

Some documents use the percentage sign % for "syllable" instead of the Sigma σ, as it is easier to type. This here document will keep using σ, but % is perfectly valid.

You can also use several of those symbols together, to fabricate more heavily conditioned rules:

$A \rightarrow B / \sigma_ \#$
 $A \rightarrow B / \#_ \#$
 $A \rightarrow B / \sigma_ \sigma] \ || \ \sigma_ \$$
 $A \rightarrow B / [' \sigma_$

Here, the first rule means that for the change to occur, the input must be positioned at the end of a word, but also be preceded by a syllable within that word. This means it does not occur in monosyllabic words, or at the first syllable of one.

On the contrary, the second indicates that the change only occurs between word boundaries, and where the only content is the one described in the rule (which by necessity means it can only happen in monosyllabic words).

The third rule, notated here in two different ways, means that the input must be the sole component of its own syllable, as it has to occur after a syllable, but also be the coda of its own syllable.

The fourth one only occurs at the beginning of stressed syllables, and makes use of the IPA's primary stress mark ' .

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Less often used symbols, because their components are less often relevant to sound changes, are $\|$ (phrase boundary or pause in speech) and $+$ (morpheme boundary). They are used in much the same ways as the ones mentioned above.

Categories

You can express that a change occurs in the presence of multiple elements, generally phonetically related in some way. For this, you must specify what your tokens mean.

Note that C and V are very often used to mean "any consonant" and "any vowel" and re-defining them might prove to be a hindrance to the reader's comprehension.

For instance, let's consider an imaginary inventory that we can break down into categories:

Label	Contents
N	mn
P	ptkbgd
F	fsxvzy
J	ljw
V	ieæuoɔ
U	uoɔ
Y	ieæ

Above, we have defined seven groups, each designated by a different token/placeholder. Those groups are commonly referred to "phonetic classes".

Each of those classes can now be used to define a specific environment:

```
A → B / _N
A → B / J_
Y → U / _C
P → F / V_V
```

Here, we have four rules making use of those tokens. Here is what they mean, in order:

- "A becomes B before any of the sounds in category **N**";
- "A becomes B after any of the sounds in category **J**";
- "Any of the sounds in category **Y** becomes the corresponding sound in **U** before a Consonant";
- "Any of the sounds in P becomes the corresponding sound in F between Vowels".

But how can we know, in the case of the two last rules, which sounds correspond?

Most often, the rules will not be as vague, but one can stumble upon a fairly opaque piece from time to time.

Usually, looking at the number of sounds in each category and their order is enough to understand the rule:

- the vowels of Y and U, three in each, are ordered from high to low in both cases, and separated in front vowels (Y) and back vowels (U). It would make sense to assume that the third rule is best formulated as *"A front vowel becomes backed before a consonant"*;
- we can see that there are 6 sounds in both P and F, and that both lists are ordered the same way: unvoiced sounds, then voiced sounds, from the front of the mouth to its back. It then makes sense to assume that the fourth rule can be read as *"Plosives become fricatives of same Place of Articulation between vowels"*.

If the elements of your set are not related in a way that is conducive to grouping them together in a token, or if they are a very small number, you may prefer using curly brackets $\{\}$.

```
A      → B / _{t,k}
A      → B / {i,e}_
{k,g} → G / _V
```

There is also the symbol for empty sets $\emptyset$:

```
 $\emptyset$  → ə / {p,t,k}_ {p,t,k}
i →  $\emptyset$  / j_#
```

The first rule is a way to write a phenomenon called **epenthesis**, the insertion of a sound, and here can be read as *"A schwa (ə) is inserted between any two of p, t, k."*

The second one represents **elision**, the deletion of a sound, and here can be read as *"i is deleted if it follows the sound j, word-finally"*.

Difference within a category

In order to express that two similar (but not identical) sounds in a rule are supposed to be different, we use subscript numbers next to them:

```
V1V2 → V1 :
A      → B / C1C2
```

This rule means that *if two vowels are next to each other, they become an elongated version of the first vowel*. The second one can be read as *A becomes B between two different consonants*.

Optional elements

Sometimes, your rules will have optional components.

```
A → B / P(J)_  
A → B / σ(C)_$  
A → B / _(XY)Z
```

The first rule indicates that *"A becomes B after a plosive (P) whether an approximant (J) is present between the input and P or not"*.

The second one would be read as *"A turns into B after a syllable, whether a C is present or not, if it (the input) occurs at the end of a syllable"*.

The third one is *"A turns into B before Z or before XYZ"*.

Parentheses encase elements that may be present but are not required. An alternative reading of one of the rules above would be, for example, *"A turns into B before Z ignoring XY"*. here, the parentheses serve to group the elements that are optional.

Using features

Let's consider once again our categories from the previous section *Categories* and how they could also be expressed using features:

- N contains two consonants of similar voicing (voiced) and manner of articulation (nasal), but different place of articulation;
- P contains consonants of dissimilar voicing and places of articulation, but similar manner of articulation (plosive);
- F contains consonants of dissimilar voicing and places of articulation, but similar manner of articulation (fricative);
- J contains consonants of similar voicing (voiced) and manner of articulation (approximant), but dissimilar places of articulation;
- V contains vowels of different heights, but that are all to the front;
- U contains vowels of different heights, but that are all to the back.

The most consistent criterion for differentiating the consonants being manner of articulation, we can deduce it is best to use that to differentiate the groups.

Here are some alternative notations, which can be used instead of defining whole groups sound by sound:

Label	Contents	Features
N	mn	[+ consonantal + nasal]
P	ptkbg	[+ consonantal -continuant -sonorant]
F	fsxvzy	[+ consonantal + continuant -sonorant]
J	ljw	[+ consonantal + continuant + sonorant]
V	ieæ	[-consonantal + sonorant + syllabic -back]
U	uoɔ	[-consonantal + sonorant + syllabic + back]

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Distinctive features can be used in phonological rules, instead of using symbols. This helps with not having a tonne of symbols to memorise definitions for, especially when several criteria are involved with each rule.

For example, instead of writing a rule like $i, u \rightarrow \tilde{i}, \tilde{u} / _ \{m, n, \eta\}$, we can abridge it slightly by calling upon the *nasal* quality of the three consonants and write $i, u \rightarrow \tilde{i}, \tilde{u} / _ [+nas -cont]$. However, both of /i u/ are high vowels. We can thus write $[+syll +high] \rightarrow [+nasal] / _ [+nasal -cont]$.

$$\begin{bmatrix} +syll \\ +high \end{bmatrix} \rightarrow [+nasal] / \text{---} \begin{bmatrix} +nasal \\ -cont \end{bmatrix}$$

This last notation still is omitting some information, but that information would be redundant. For example's sake, though, here is a more complete notation that is equally valid:

$$\begin{bmatrix} +syll \\ +high \\ -low \end{bmatrix} \rightarrow \begin{bmatrix} +syll \\ +high \\ -low \\ +nasal \end{bmatrix} / \text{---} \begin{bmatrix} +nasal \\ -cont \end{bmatrix}$$

In this latest example, we are making the unnecessary specification that what is *high* is not also *low*, that the *nasal* feature is added to the 3 previous ones.

When conditioning sound changes, distinctive features can be used in conjunction with *variables*, standing for their binary values (+ and -).

$$[+nasal] \rightarrow \begin{bmatrix} +ant \\ +cor \end{bmatrix} / \text{---} \begin{bmatrix} +ant \\ +cor \end{bmatrix}$$

$$[+nasal] \rightarrow \begin{bmatrix} +ant \\ -cor \end{bmatrix} / \text{---} \begin{bmatrix} +ant \\ -cor \end{bmatrix}$$

The two rules above mean, respectively:

1. "A nasal becomes [+anterior +coronal] if the sound after it is [+anterior +coronal]";
2. "A nasal becomes [+anterior -coronal] if the sound after it is [+anterior -coronal]";

However, these rules could be summarised as follows, using the Greek letters Alpha α and Beta β as variables:

$$[+nasal] \rightarrow \begin{bmatrix} \alpha ant \\ \beta cor \end{bmatrix} / \text{---} \begin{bmatrix} \alpha ant \\ \beta cor \end{bmatrix}$$

"A nasal becomes of same anteriority and coronality as the sound that follows it".

We could also rule that the nasal becomes of opposite anteriority and coronality as the sound that follows it with:

$$[+nasal] \rightarrow \begin{bmatrix} \alpha_{ant} \\ \beta_{cor} \end{bmatrix} / \text{---} \begin{bmatrix} \beta_{ant} \\ \alpha_{cor} \end{bmatrix}$$

Here, we use the letters Alpha and Beta because they are the first two, and we only need to represent two values: $+$ and $-$.

The same letter, Alpha α or Beta β , being used on both sides of a rule signifies that the feature maintains its value of $+$ or $-$, while different signs mean it changes its value from $+$ to $-$, or $-$ to $+$.